Electronic Journal of Statistics
ISSN: 1935-7524

Doubly-robust sufficient variable selection in single-index models with outlier contamination

Abdul-Nasah Soale[1], **Adewale F. Lukman**[2] **and Essoham Ali**[3,4]

[1]*Department of Mathematics, Applied Mathematics, and Statistics Case Western Reserve University, Cleveland, OH, USA., e-mail:* abdul-nasah.soale@case.edu

[2]*Department of Mathematics and Statistics University of North Dakota, Grand Forks, ND 58202, USA , e-mail:* adewale.lukman@und.edu

[3]*CNRS UMR 6205, Univ Bretagne Sud Vannes, France. , e-mail:* essoham.ali@univ-ubs.fr

[4]*Institut de Mathématiques Appliquées, UCO Angers 49000, France.,*

Abstract: Case-influence diagnostics play a fundamental role in regression analysis, yet they are seldom incorporated into modern variable selection procedures, particularly in high-dimensional settings where penalized estimators can be highly sensitive to a small number of atypical observations. We propose a new method, termed sparse outer product of gradients based on local median regression (SOPG-LAD), for variable selection in single-index models with influential observations. The method proceeds in two stages. First, ℓ_1-penalized locally weighted median regression is used to obtain sparse coefficient estimates that attenuate the effects of leverage points and response outliers. These local estimates are then aggregated to recover the global index structure and identify relevant predictors through a regularized singular value decomposition. This construction imposes sparsity at both the local and global levels, yielding robustness to multiple forms of contamination. In contrast to existing approaches, SOPG-LAD explicitly accounts for influential observations during both estimation and selection. As a result, it provides improved stability and more reliable sparsity recovery under heavy-tailed errors and leverage contamination. Simulation studies covering a wide range of dimensional regimes and contamination mechanisms demonstrate substantial gains over classical penalized methods. Applications to air pollution analysis and gene selection further illustrate the practical relevance of the proposed procedure. We also establish the consistency of the estimator.

MSC2020 subject classifications: Primary 62F07, 62J20; secondary 62B05.
Keywords and phrases: influential observations, leverage points, variable selection, sufficient dimension reduction, sparse local median estimates.

1. Introduction

Single-index models constitute a fundamental class of semiparametric regression methods and have found wide application in economics, epidemiology, and biomedical research [12, 30]. By combining dimension reduction with flexible link functions, these models offer an appealing compromise between interpretability and modeling flexibility. Nevertheless, their practical performance can be substantially undermined by atypical observations, which are prevalent in modern, high-dimensional datasets [24, 26]. Observations that exert disproportionate influence on parameter estimates - commonly referred to as *influential points* - have therefore attracted sustained attention in the literature [6, 22, 27].

The sensitivity of single-index estimators to data contamination depends critically on the underlying estimation strategy. Classical approaches based on ordinary least squares [12, 19]

and related mean-based dimension reduction techniques, including sliced inverse regression [18], minimum average variance estimation [33] and their extensions, are known to be vulnerable to heavy-tailed errors and high-leverage observations. These limitations have motivated the development of robust alternatives built on quantile and rank-based principles [14, 10, 15, 4], which improve resistance to response contamination and distributional misspecification.

Robustness considerations also play a central role in variable selection for single-index models. Existing studies have demonstrated that the accuracy of sparsity recovery is closely tied to the estimation framework [14, 20, 4, 28]. However, the impact of contamination on variable selection exhibits important subtleties. When influential observations arise primarily from leverage effects, local smoothing procedures can mitigate their influence, largely independent of the regression criterion. In such settings, sparse minimum average variance estimation [32] may perform comparably to robust penalized estimators such as the penalized composite quantile regression method of [28] or the penalized generalized signed-rank approach of [4].

In contrast, when contamination is driven by heavy-tailed or asymmetric errors, the choice of regression functional becomes decisive. Composite quantile procedures may accumulate noise across multiple quantile levels, leading to inflated false selections, while rank-based methods are sensitive to distortions in the ordering of observations. Although the choice of penalty function also influences selection performance, this work focuses on widely adopted sparsity-inducing penalties, including the least absolute shrinkage and selection operator [31], the smoothly clipped absolute deviation penalty [8], and the minimax concave penalty [36].

Motivated by these considerations, we propose a robust and scalable variable selection framework based on sparse outer products of gradients constructed from locally weighted least absolute deviation regression. The resulting procedure, termed *SOPG-LAD*, integrates the stability of local smoothing with the robustness of the least absolute deviation (median regression). By imposing sparsity at both the local estimation stage and the global gradient aggregation stage, the method effectively suppresses covariates with negligible contributions, thereby reducing the risk of spurious discoveries.

The proposed framework is well-suited for both moderate and ultra-high-dimensional settings and naturally supports sequential screening and refinement of predictors. This enables practitioners to evaluate the marginal and joint relevance of covariates in a data-adaptive manner. Extensive theoretical analysis and numerical studies demonstrate that SOPG-LAD achieves improved robustness and selection accuracy under a broad range of contamination mechanisms, providing a unified tool for reliable dimension reduction and variable selection in single-index models.

The remainder of this article is organized as follows. The setup and model formulation are given in Section 2. In Section 3, we introduce the variable selection estimator based on sparse outer product of gradients of sparse local median regression (SOPG-LAD). This is followed by the finite sample estimation, implementation issue, and theoretical properties in Section 4. We provide extensive simulation studies under a spectrum of contamination mechanisms including outliers, collinearity, and different covariate dimensionality in Section 5. Two real applications on air pollution in US cities and riboflavin production in *Bacillus subtilis* datasets are presented in Section 6. Concluding remarks are offered in Section 7. All further results are relegated to the appendix.

Notation. Throughout the paper, we utilize the following notations. We use $\|.\|$ to denote the ℓ_2-norm, $\|.\|_1$ to denote the ℓ_1-norm, and $\|.\|_0$ to denote the number of nonzero elements. We denote the cardinality of a set A by $|A|$, and sample mean by $\mathbb{E}_n(.)$, i.e, $\mathbb{E}(\boldsymbol{x}) = n^{-1}\sum_{i=1}^{n}\boldsymbol{x}_i$. Lastly, for constants a and b, we use $a \vee b = \max\{a, b\}$ and $a \wedge b = \min\{a, b\}$.

2. Setup and Model

Let $\mathbf{X} \in \mathbb{R}^p$ denote a random vector of covariates and let $Y \in \mathbb{R}$ be a scalar response. We consider the following general single-index model,

$$Y = g(\boldsymbol{\beta}^\top \mathbf{X}, \epsilon), \tag{1}$$

where $g(\cdot,\cdot)$ is an unknown link function, $\boldsymbol{\beta} \in \mathbb{R}^p$ is an unknown parameter vector satisfying $\|\boldsymbol{\beta}\| = 1$ for identifiability, and ϵ is a random error term such that $\mathbb{E}(\epsilon \mid \mathbf{X}) = 0$ almost surely. This conditional mean restriction allows the distribution of ϵ to depend on $\mathbf{X}$.

Model (1) includes the classical single-index model

$$Y = g(\boldsymbol{\beta}^\top \mathbf{X}) + \epsilon,$$

as a special case, and also accommodates heteroscedastic structures of the form

$$Y = g_1(\boldsymbol{\beta}^\top \mathbf{X}) + g_2(\boldsymbol{\beta}^\top \mathbf{X})\,\epsilon,$$

where $g_1(\cdot)$ and $g_2(\cdot)$ are unknown smooth functions.

Regardless of the specific form of (1), the response Y depends on the covariates $\mathbf{X}$ only through the single index $\boldsymbol{\beta}^\top \mathbf{X}$. Consequently, the conditional independence relation

$$Y \perp\!\!\!\perp \mathbf{X} \mid \boldsymbol{\beta}^\top \mathbf{X} \tag{2}$$

holds, where $\perp\!\!\!\perp$ denotes statistical independence.

Since any nonzero scalar multiple of $\boldsymbol{\beta}$ satisfies (2), the parameter vector itself is identifiable only up to scale. We therefore focus on the one-dimensional subspace $\mathrm{span}(\boldsymbol{\beta})$, which has been shown to be identifiable under mild regularity conditions [34]. This subspace, defined as the column space of $\boldsymbol{\beta}$, is commonly referred to as the *central subspace* for the regression of Y on $\mathbf{X}$ and is typically denoted by $\mathcal{S}_{Y|X}$.

Within the sufficient dimension reduction literature, a variety of model-free procedures have been developed to estimate $\boldsymbol{\beta}$ under weak distributional assumptions; see, for example, [18], [33], and [16]. In contrast to classical sufficient dimension reduction, which aims to recover linear combinations of all predictors that preserve regression information, our primary objective is variable selection. Specifically, we seek to identify the smallest subset of covariates that retains essentially all information about the conditional distribution of Y given $\mathbf{X}$.

Following (2), if the basis $\boldsymbol{\beta}$ contains zeroes or coordinates arbitrarily close to zero, then model (2) can be satisfied without those corresponding covariates. More formally, suppose we partition $\boldsymbol{\beta}$ as

$$\boldsymbol{\beta} = \begin{pmatrix} \boldsymbol{\beta}_{\mathcal{A}} \\ \boldsymbol{\beta}_{\mathcal{A}^c}, \end{pmatrix}, \text{ where } \begin{cases} \boldsymbol{\beta}_{\mathcal{A}} \in \mathbb{R}^{|\mathcal{A}|}, \\ \boldsymbol{\beta}_{\mathcal{A}^c} \in \mathbb{R}^{p-|\mathcal{A}|}, \end{cases} \tag{3}$$

where $\mathcal{A}$ is the smallest set of indices of the active covariates and $\mathcal{A}^c = \{1, \ldots, p\} \backslash \mathcal{A}$ with

$$\boldsymbol{\beta}_{\mathcal{A}} : \{\beta_j \neq 0,\ j \in \mathcal{A}\} \text{ and } \boldsymbol{\beta}_{\mathcal{A}^c} : \{\beta_{j'} \approx 0,\ j' \in \mathcal{A}^c\}.$$

Accordingly, if we partition $\mathbf{X} = (\mathbf{X}_{\mathcal{A}}^\top, \mathbf{X}_{\mathcal{A}^c}^\top)^\top$, then we can re-expressed (2) as

$$Y \perp\!\!\!\perp \mathbf{X}_{\mathcal{A}^c}^\top \mid \mathbf{X}_{\mathcal{A}}^\top \text{ approximately}, \tag{4}$$

where $\mathbf{X}_{\mathcal{A}}^\top$ contains the smallest set of active variables in $\mathbf{X}$ required to predict Y without loss of information. In what follows, we will focus on determining $\mathbf{X}_{\mathcal{A}}^\top$ through $\boldsymbol{\beta}_{\mathcal{A}}$.

3. Active set selection via SOPG-LAD

By the definition in (2), we also have

$$F_{Y|\mathbf{X}}(y \mid \boldsymbol{x}) = F_{Y|\boldsymbol{\beta}^\top \mathbf{X}}(y \mid \boldsymbol{\beta}^\top \boldsymbol{x}), \tag{5}$$

where $F_{Y|\mathbf{X}}(y \mid \boldsymbol{x}) = \Pr(Y \leq y \mid \mathbf{X} = \boldsymbol{x})$ denotes the conditional distribution function of Y given $\mathbf{X} = \boldsymbol{x}$. Thus, for any non-degenerate Y, the corresponding conditional quantiles

$$Q_\tau(Y \mid \mathbf{X} = \boldsymbol{x}) = Q_\tau(Y \mid \boldsymbol{\beta}^\top \mathbf{X} = \boldsymbol{\beta}^\top \boldsymbol{x}), \text{ for all } \tau \in (0, 1), \tag{6}$$

where $Q_\tau(Y \mid \mathbf{X} = \boldsymbol{x}) = \inf\{y : F_{Y|\mathbf{X}}(y \mid \boldsymbol{x}) \geq \tau\}$. This implies we can obtain information about the entire conditional distribution of Y given $\mathbf{X}$ by examining $Q_\tau(Y \mid \mathbf{X})$ across all values of τ. Lemma 1 in [13] establishes the equivalence between $\mathcal{S}_{Y|\mathbf{X}}$ based on the original response and the conditional quantiles. For ease of reference, we restate it here.

Proposition 3.1 (Lemma 1 of [13])**.** *For non-degenerate* $\mathbf{X}$ *and* Y*, we have* $\mathcal{S}_{Y|\mathbf{X}} = \mathcal{S}_{Q_\mathbf{X}|\mathbf{X}}$*, where* $Q_\mathbf{X}(.) = F_{Y|\mathbf{X}}^{-1}(.)$ *denotes the stacked quantiles for all* $\tau \in (0, 1)$.

Proposition 3.1 is also equivalent to Lemma 1 in [15], where the proof is provided.

Now, suppose $F_{Y|\mathbf{X}}$ is differentiable at any given $\mathbf{X} = \boldsymbol{x}$. Then following [33], we can find directions in the central space from the gradients of $Q_\tau(Y \mid \mathbf{X} = \boldsymbol{x})$ at any given τ level.

Proposition 3.2. *Assume* $F_{Y|\mathbf{X}}$ *is differentiable at* $\mathbf{X} = \boldsymbol{x}$*. Then the gradient direction* $\dfrac{\partial Q_\tau(Y \mid \mathbf{X} = \boldsymbol{x})}{\partial \boldsymbol{x}} \in \mathcal{S}_{Y|X}$.

By Propositions 3.1 and 3.2, the central space and subsequently the active set can be determined by aggregating the outer product of the gradients across τ as in [15].

However, here we deviate from the composite quantile idea and follow in the spirit of non-parametric approximation of $\mathbb{E}(Y|\mathbf{X})$. Notice that if $F_{Y|\mathbf{X}}$ is differentiable at any given point $\mathbf{X} = \boldsymbol{x}$, then the first-order Taylor expansion of $Q_\tau(Y \mid \mathbf{X} = \boldsymbol{x})$ around $\mathbf{X} = \boldsymbol{x}_0$ is given by

$$Q_\tau(Y \mid \mathbf{X} = \boldsymbol{x}) \approx Q_\tau(Y \mid \boldsymbol{x}_0) + \left[\frac{\partial Q_\tau(Y \mid \boldsymbol{x})}{\partial \boldsymbol{x}}\right](\mathbf{X} - \boldsymbol{x}_0).$$

If this expansion holds, then we should be able to estimate the sparse central space from the outer-product of the gradients of $Q_\tau(Y \mid \mathbf{X} = \boldsymbol{x})$ and consequently, perform variable selection. Note that because the differentiability of $Q_\tau(Y \mid \boldsymbol{\beta}^\top \boldsymbol{x})$ is not always guaranteed, we assume

Y is continuous with a strict monotone density. Also, while considering gradients across different quantile levels gives a better picture of the conditional distribution, the gradients at some levels of τ may not be informative about the central space although they may be non-zero due to outliers. Therefore, [15] suggested using an adaptively weighted outer-product, which does not control for spurious contributions due to a few influential observations.

Since, we are interested in variable selection, excluding uninformative quantiles automatically gets us closer to our desired goal. And as the median is more informative about the center of the distribution than any other quantile, we only focus on the local median, i.e., $\tau = 0.5$. Therefore, we approximate $Q(Y|\mathbf{X} = \boldsymbol{x}_0)$ as

$$Q(Y|\mathbf{X} = \boldsymbol{x}_0) \approx Q_{0.5}(Y|\mathbf{X} = \boldsymbol{x}_0) = a + \mathbf{b}^\top(\mathbf{X} - \boldsymbol{x}_0),$$

where $a \in \mathbb{R}$ and $\mathbf{b} \in \mathbb{R}^p$. Define positive weights

$$w(\boldsymbol{x}_0) = \frac{K_h(\mathbf{X} - \boldsymbol{x}_0)}{\mathbb{E}[K_h(\mathbf{X} - \boldsymbol{x}_0)]},$$

where $K_h(.)$ is a kernel density function with bandwidth $h > 0$. We estimate $(a, \mathbf{b})$ at $\mathbf{X} = \boldsymbol{x}_0$ by minimizing the penalized weighted least absolute deviation

$$\left(\hat{a}(\boldsymbol{x}_0), \hat{\mathbf{b}}(\boldsymbol{x}_0)\right) = \underset{a\in\mathbb{R},\ \mathbf{b}\in\mathbb{R}^p}{\arg\min}\ \mathbb{E}\left[w(\boldsymbol{x}_0)|Y - a - \mathbf{b}^\top(\mathbf{X} - \boldsymbol{x}_0)|\right] + P_\lambda(\mathbf{b}), \tag{7}$$

where $P_\lambda(\mathbf{b})$ is a sparsity-inducing penalty term with tuning parameter $\lambda > 0$.

To dampen the influence of high-leverage points, we choose the kernel to down-weight points farther from x_0. Thus, we adopt the Gaussian radial basis function for the kernel, given by

$$K_h(u) = \frac{1}{h\sqrt{2\pi}} \exp\left(-\frac{u^2}{2h^2}\right),$$

where the tuning parameter h controls the kernel radius. While the choice of $P_\lambda(\mathbf{b})$ is flexible, we only focus on the ℓ_1 penalty, i.e., $P_\lambda(\mathbf{b}) = \lambda \sum_{k=1}^p |b_k|$, which is readily available in several statistical software. Moreover, using the ℓ_1 penalty reduces (7) to a convex optimization, which guarantees good asymptotic properties.

Denote the outer product of gradients as

$$\boldsymbol{\Lambda} = \mathbb{E}\left(\frac{\partial Q_{0.5}(Y \mid \boldsymbol{x})}{\partial \boldsymbol{x}} \frac{\partial Q_{0.5}(Y \mid \boldsymbol{x})}{\partial \boldsymbol{x}^\top}\right).$$

Notice that $\boldsymbol{\Lambda} = \mathbb{E}\left(\frac{\partial Q_{0.5}(Y \mid \boldsymbol{\beta}^\top \boldsymbol{x})}{\partial \boldsymbol{\beta}^\top \boldsymbol{x}}\right)^2 \boldsymbol{\beta}\boldsymbol{\beta}^\top$ is rank 1 with only one nonzero eigenvalue whose corresponding eigenvector is $\boldsymbol{\beta}$. Let $\mathbf{B} = \{\mathbf{b}(\boldsymbol{x}_0) : \boldsymbol{x}_0 \in \mathbf{X}\}$ denote a column-wise stacked vector of local gradients. If there exists contamination in the data, the few irrelevant variables selected at some local points are expected to correspond to sparse columns of $\mathbf{B}$. Thus, instead of simply performing eigen decomposition on $\boldsymbol{\Lambda}$, we impose sparsity on the loadings to force such sparse columns to zero.

One issue that may arising is determining the level of sparsity to impose on the loadings of $\boldsymbol{\Lambda}$ that will not lead to significant loss of the true active covariates. To address this, we

propose a contribution criterion to determine which variable gets selected into the active set. This is particularly important in real applications, where the investigator is unsure how many variables are relevant, or in scenarios where the investigator wants to rank variables based on their contribution. To achieve this, we start with one variable and incrementally increase the number of non-zero loadings based on the relative change in the leading singular value until the change falls below a user-specified threshold, which can be adjusted depending on the desired level of parsimony.

4. Finite sample estimation

We next describe the SOPG-LAD variable selection based on the random sample $(\mathbf{X}_n, Y_n) = \{(\boldsymbol{x}_1, y_1), \ldots, (\boldsymbol{x}_n, y_n)\}$ of $(\mathbf{X}, Y)$.

4.1. Active set estimation

Define the weight function for the neighborhood of x_i as

$$w_{ij} = \frac{\exp\left(-\gamma^2 \|\boldsymbol{x}_j - \boldsymbol{x}_i\|\right)}{\sum_{j=1}^{n} \exp\left(-\gamma^2 \|\boldsymbol{x}_j - \boldsymbol{x}_i\|\right)}, \qquad j = 1, \ldots, n, \tag{8}$$

and solve the penalized weighted local median regression

$$(\hat{a}_i, \hat{\mathbf{b}}_i)^\top = \arg\min_{a,\mathbf{b}} \frac{1}{n} \sum_{j=1}^{n} w_{ij} \left| y_j - a - \mathbf{b}^\top (\boldsymbol{x}_j - \boldsymbol{x}_i) \right| + \lambda \|\mathbf{b}\|_1, \tag{9}$$

where $\lambda > 0$. To ensure fair penalization, $\mathbf{X}_n$ is standardized so that each covariate has a zero mean and unit variance.

Let $\tilde{\mathbf{B}} = (\hat{\mathbf{b}}_1, \ldots, \hat{\mathbf{b}}_n)^\top \in \mathbb{R}^{n \times p}$. We perform sparse principal component analysis on $\tilde{\mathbf{B}}$ via regularized singular value decomposition (sPCA-rSVD), details of which can be found in [25]. If $\operatorname{rank}(\tilde{\mathbf{B}}) = r$, the SVD of $\tilde{\mathbf{B}}$ admits the decomposition

$$\tilde{\mathbf{B}} = \mathbf{U}\mathbf{D}\mathbf{V}^\top, \text{ where } \mathbf{U} = (\boldsymbol{u}_1, \ldots, \boldsymbol{u}_r) \in \mathbb{R}^{n \times r} \text{ and } \mathbf{V}^\top = (\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p) \in \mathbb{R}^{r \times p}$$

are left and right singular vectors with orthonormal columns, respectively. The diagonal matrix $\mathbf{D} = \operatorname{diag}(d_1, \ldots, d_r) \in \mathbb{R}^{r \times r}$, where $d_1 \geq d_2 \geq \cdots \geq d_r > 0$ are the singular values. The columns of $\mathbf{UD}$ are the principal components with corresponding loadings as the columns of $\mathbf{V}$.

For our case, $r = 1$. Thus, we focus on obtaining the best rank-one approximation of $\tilde{\mathbf{B}}$ using sPCA-rSVD. To that end, we set $\boldsymbol{\eta} = d_1 \boldsymbol{v}_1$ and solve the optimization problem

$$\min_{\boldsymbol{\eta}, \boldsymbol{u}_1} \left\| \tilde{\mathbf{B}} - \boldsymbol{u}_1 \boldsymbol{\eta}^\top \right\|_F^2 + P_\xi(\boldsymbol{\eta}) \quad \text{subject to } \|\boldsymbol{u}_1\| = 1, \tag{10}$$

where $\|\cdot\|_F$ denotes the Frobenius norm, $P_\xi(\boldsymbol{\eta})$ is a sparsity-inducing penalty with a tuning parameter $\xi > 0$. The non-zero entries of $\boldsymbol{\eta}$ identify the selected variables. The magnitude of the loadings provide a natural ranking of variable importance: larger absolute values correspond to more influential covariates. We summarize the estimation in Algorithm 1.

4.2. Implementation Issues

For the Gaussian radial kernel weights in Step 2, we follow the recommendation of [17] and choose

$$\hat{\gamma} = \binom{n}{2}^{-1} \sum_{1 \le i < j \le n} \|\boldsymbol{x}_i - \boldsymbol{x}_j\|. \tag{11}$$

The median regression coefficients in Step 4 can be computed using existing statistical software. In our implementation, we employ the `conquer` package in R for penalized quantile regression. In Step 7, the leading sparse rank-one approximation of $\tilde{\mathbf{B}}$ is obtained using the `ssvd` function from the `irlba` package. The regularization parameters $\lambda \in (0, 1)$ and ξ are selected by cross-validation over prespecified grids of candidate values. This tuning procedure can be efficiently carried out using the computational routines provided in the `conquer` and `irlba` packages.

The desired number of active variables $s = \|v_1\|_0$ can be specified directly by the user. Alternatively, an automatic selection rule for s could be used when the investigator is unsure how many variables are relevant, which is usually the case in practice. The threshold for contribution depends on the investigators preference.

Algorithm 1 SOPG-LAD Procedure

1: **for** $i = 1, \ldots, n$ **do**
2: Compute kernel weights

$$\hat{w}_{ij} = \frac{\exp\left(-\|\boldsymbol{x}_i - \boldsymbol{x}_j\|/\hat{\gamma}^2\right)}{\sum_{k=1}^{n} \exp\left(-\|\boldsymbol{x}_i - \boldsymbol{x}_k\|/\hat{\gamma}^2\right)}, \qquad j = 1, \ldots, n.$$

3: Set $\tilde{Y}_n = \sqrt{\hat{w}_i}\, Y_n$ and $\tilde{\mathbf{X}}_n = \sqrt{\hat{w}_i}\,(\mathbf{X}_n - \boldsymbol{x}_i)$, where $\hat{w}_i = (\hat{w}_{i1}, \ldots, \hat{w}_{in})^\top$.
4: Compute the LASSO median regression estimate $\hat{\mathbf{b}}_i$ for regressing $\tilde{Y}_n$ on $\tilde{\mathbf{X}}_n$.
5: **end for**
6: Form the matrix of local gradients $\tilde{\mathbf{B}}_n = (\hat{\mathbf{b}}_1, \ldots, \hat{\mathbf{b}}_n)^\top$.
7: Choose s (the desired number of non-zero loadings in $\boldsymbol{v}_1$) and compute $(\hat{\boldsymbol{u}}_1, \hat{\boldsymbol{v}}_1, \hat{d}_1) = \texttt{ssvd}(\tilde{\mathbf{B}}_n, k = 1, n = s)$.
8: **if** s is not specified **then**
9: Initialize $s_0 \leftarrow 1$ and set the minimum contribution threshold $\varrho \leftarrow 0.005$ (i.e., 0.5%).
10: **while** TRUE **do**
11: $s_0 \leftarrow s_0 + 1$ and recompute $\hat{d}_1^{\text{update}}$ from $\texttt{ssvd}(\tilde{\mathbf{B}}_n, k = 1, n = s_0)$.
12: $\Delta \leftarrow (\hat{d}_1^{\text{update}} - \hat{d}_1)/\hat{d}_1$.
13: **if** $s_0 > p$ **or** $\Delta < \varrho$ **then**
14: **break**
15: **end if**
16: **end while**
17: **end if**

4.3. Theoretical Properties

Let $\boldsymbol{\theta} = (a, \mathbf{b}^\top)^\top$ and $\mathbf{Z} = (\mathbf{X}^\top, Y)^\top$. Define the objective function

$$\mathcal{L}(\boldsymbol{\theta}, \mathbf{Z}; \boldsymbol{x}_0) = \mathbb{E}\left[w(\boldsymbol{x}_0)|Y - a - \mathbf{b}^\top(\mathbf{X} - \boldsymbol{x}_0)|\right] + \lambda\|\mathbf{b}\|_1. \tag{12}$$

Notice that both the absolute loss $|Y - a - \mathbf{b}^\top(\mathbf{X} - \boldsymbol{x}_0)|$ and the penalty term $\|\mathbf{b}\|_1$ are convex functions of $\mathbf{b}$ but not differentiable everywhere as $|b_j|$ is not differentiable at $b_j = 0$. By the property of convexity, $w(\boldsymbol{x}_0)|Y - a - \mathbf{b}^\top(\mathbf{X} - \boldsymbol{x}_0)|$ remains convex. Therefore, at any given $\mathbf{X} = \boldsymbol{x}_0$, $\mathcal{L}(\boldsymbol{\theta}, \mathbf{Z}; \boldsymbol{x}_0)$ is convex, which makes the minimizer, i.e., $\arg\min_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}, \mathbf{Z}; \boldsymbol{x}_0)$ an M-estimator. Thus, we can leverage properties of M-estimators to establish the asymptotic properties based on the following standard assumptions.

(A1) The covariate $\mathbf{X} = (X_1, \ldots, X_p)^\top$ is normalized such that $\mathbb{E}(X_i) = 0$ and $\mathbb{E}(X_i^2) = 1$, $i = 1, \ldots, p$.

(A2) The support of $\mathbf{X}$ denoted $\mathrm{Supp}(\mathbf{X})$ is a compact and convex subset of $\subseteq \mathbb{R}^p$. For each $\boldsymbol{x} \in \mathrm{Supp}(\mathbf{X})$, the conditional density $f_{Y|\mathbf{X}}(y|\boldsymbol{x})$ is continuously differentiable with respect to y. We further assume $f_{Y|\mathbf{X}}(\boldsymbol{\beta}^\top \boldsymbol{x}|\boldsymbol{x}) > 0$ for each $\boldsymbol{x} \in \mathrm{Supp}(\mathbf{X})$. Lastly, we assume both $f_{Y|\mathbf{X}}(y|\boldsymbol{x})$ and $f'_{Y|\mathbf{X}}(y|\boldsymbol{x})$ are absolutely bounded uniformly in y for each $\boldsymbol{x} \in \mathrm{Supp}(\mathbf{X})$.

(A3) The bandwidth $h = o(1)$ and $nh^p/C_n \to \infty$, where $C_n = (\log n)^{C'}$ for some arbitrary constant $C' > 0$.

(A4) At each $\boldsymbol{x}_0 \in \mathbf{X}$, $\|\mathbf{b}(\boldsymbol{x}_0)\|_0 \le s$, where $s = o(n)$ is the number of non-zero coefficients in $\mathbf{b}(\boldsymbol{x}_0)$.

(A5) The gram matrix $\mathbb{E}(\boldsymbol{x}_i \boldsymbol{x}_i^\top)$ is such that

$$\inf_{\boldsymbol{\delta} \in \mathbb{R}^p : \boldsymbol{\delta} \neq 0} \frac{\boldsymbol{\delta}^\top \mathbb{E}(\boldsymbol{x}_i \boldsymbol{x}_i^\top) \boldsymbol{\delta}}{\|\boldsymbol{\delta}\|^2} > 0.$$

Assumption (A1) ensures that the covariates are well-behaved to allow for fair penalization. The second assumption (A2) imposes mild smoothness on the density, which is standard in for conditional quantile estimators [21, 2]. Assumption (A3) is needed to ensure consistency in the local estimation and it is standard for local smooth estimators [9]. The sparsity assumption in (A4) ensures that for some appropriate regularizing parameter λ, there exists penalized estimators at any given $\boldsymbol{x}_0 \in \mathbf{X}$ with some coordinates of $\mathbf{b}(\boldsymbol{x}_0) = 0$ [2]. Lastly, assumption (A5) is a weaker assumption that replaces the stronger requirement for $\mathbb{E}(\boldsymbol{x}_i \boldsymbol{x}_i^\top)$ to be positive definite which does not hold when $p > n$ and even in low-rank designs with $p < n$. This assumption is also standard for Lasso and other penalized selectors [3].

Lemma 4.1. *Let the parameter space $\boldsymbol{\Theta}$ be a compact subset of $\mathbb{R}^{p+1}$. For all $\boldsymbol{x}_0, \boldsymbol{x}_i \in \mathbf{X}$, consider the set of weights $w(\boldsymbol{x}_0) = \{w_i\}$, where $w_i > 0$ if $\boldsymbol{x}_i \neq \boldsymbol{x}_0$ and $w_i = 0$ if $\boldsymbol{x}_i = \boldsymbol{x}_0$ with $\sum_{\forall \boldsymbol{x}_i} w(\boldsymbol{x}_0) = 1$. We have*

$$\hat{\mathbf{b}}(\boldsymbol{x}_0) \xrightarrow{p} \mathbf{b}_0(\boldsymbol{x}_0) \ \ \textit{as } n \to \infty,$$

where $\mathbf{b}_0(\boldsymbol{x}_0)$ is the true parameter at $\mathbf{X} = \boldsymbol{x}_0$.

For a general local linear M-estimator based on a convex loss ℓ, i.e.,

$$(\hat{a}(\boldsymbol{x}_i), \hat{\mathbf{b}}(\boldsymbol{x}_i))^\top = \arg\min_{a, \mathbf{b}} \frac{1}{n} \sum_{j=1}^{n} K\left(\frac{\boldsymbol{x}_j - \boldsymbol{x}_i}{h}\right) \ell[y_j - a - \mathbf{b}^\top(\boldsymbol{x}_j - \boldsymbol{x}_i)],$$

the asymptotic bias of $\hat{\mathbf{b}}(\boldsymbol{x}_i)$ is of the order h^2 and the variance is of the order $(nh^p)^{-1/2}$ under the standard regularity conditions, see [7] and [9]. As for the global ℓ_1-penalized quantile

regression, [2] showed that quantile regression estimators are consistent with a near oracle rate of $\sqrt{\frac{s \log p}{n}}$.

Following the results of [9] and [2], if conditions A1-A5 are satisfied and we choose λ adaptively based on the recommendation of [2], the respective orders of the bias and variance are

$$\text{Bias}\big(\mathbf{b}(\boldsymbol{x})\big) = O(h^2) \text{ and } \text{Var}\big(\hat{\mathbf{b}}(\boldsymbol{x})\big) = O\left(\sqrt{\frac{s \log p}{nh^p}}\right).$$

Thus, the bias neither depends on n nor p but on the smoothness of the target function. Asymptotically, the mean square error (MSE) as a function of h is given by

$$\text{MSE} \asymp h^4 + \frac{1}{nh^p}.$$

Based on this, we can determine the optimal bandwidth h_{opt}, i.e.,

$$h^4 \sim \frac{s \log p}{nh^p} \Rightarrow h_{opt} \sim \left(\frac{s \log p}{n}\right)^{1/(4+p)}.$$

Plugging h_{opt} back into MSE(h), we obtain the minimax rate of order $\left(\frac{s \log p}{n}\right)^{4/4+p}$. Therefore, the local estimator $\hat{\mathbf{b}}(\boldsymbol{x}_i)$ is consistent at this rate even if the selected model does not include some of the true covariates.

Theorem 4.2. *Assume* $\boldsymbol{\Lambda}_n = \tilde{\mathbf{B}}_n^\top \tilde{\mathbf{B}}_n$ *satisfies*

$$\boldsymbol{\Lambda}_n = \boldsymbol{\Lambda} + o_p(n^{-1/2}),$$

where $\text{span}(\boldsymbol{\Lambda}) \subseteq \mathcal{S}_{Y|\mathbf{X}}$. *If as* $n, p, s \to \infty$, $\frac{s}{n} \to o\left(\frac{1}{\log(p/s)}\right)^2$, *then the estimator* $\hat{\boldsymbol{\eta}}$ *is such that* $\Pr(\hat{\mathcal{A}} = \mathcal{A}) \to 1$.

5. Simulation Study

In this section, we evaluate the finite-sample performance of the proposed method using simulated data contaminated with outliers. SOPG-LAD is compared with several existing approaches for sufficient variable selection. Although SOPG-LAD accommodates both monotone and symmetric link functions, the simulations focus on monotone links in order to facilitate direct comparison with penalized ordinary least squares (OLS) methods, which remain widely used in practice. We also consider sparse sliced inverse regression with an ℓ_1 penalty (SIR-LASSO), which is capable of handling more general dependence structures and can be conveniently implemented using the `LassoSIR` package in R. To assess performance across different regimes, both low- and ultra-high-dimensional settings are examined. Penalized composite quantile regression was investigated using the `cqrReg` package; however,

under the configurations considered, the resulting estimates exhibited limited sparsity. In addition, sparse minimum average variance estimation is primarily designed for settings with $p < n$. For these reasons, neither PCQR nor SMAVE is included in the subsequent simulation studies.

The data generation process is as follows. Fix $(n, p) \in \{(50, 10), (50, 200)\}$ and $(n, p) \in \{(100, 10), (100, 1000)\}$, and set the proportion of contamination to $p_{\text{out}} \in \{2\%, 5\%\}$. Thus, $n_0 = n(1 - p_{\text{out}})$ of the observations are normal while the remaining $n_1 = n - n_0$ are contaminated. Next, set

$$\boldsymbol{\beta} = \begin{pmatrix} \boldsymbol{\beta}_{\mathcal{A}} \\ \boldsymbol{\beta}_{\mathcal{A}^c} \end{pmatrix}, \text{ where } \begin{cases} \boldsymbol{\beta}_{\mathcal{A}} \in \{2, -1.5, 0.8, 1.5, 3\}, \\ \boldsymbol{\beta}_{\mathcal{A}^c} = \mathbf{0} \in \mathbb{R}^{p-5}, \end{cases}$$

The coordinates of $\boldsymbol{\beta}$ are randomly shuffled so that all covariates are equally likely to be in the active set. The response is generated based on the following models under three different types of contamination.

For $i = 1, \ldots, n_0$, we generate the uncontaminated covariates as

$$\boldsymbol{x}_i \sim N_p(\mathbf{0}, \boldsymbol{\Sigma}), \text{ where } \boldsymbol{\Sigma}_{k,l} = 0.75^{|k-l|} \text{ for } k, l = 1, \ldots, p,$$

and the response based on the models:

$$\begin{aligned} \text{I} &: y_i = 0.5 + \boldsymbol{\beta}^\top \boldsymbol{x}_i + 0.8\epsilon_i, \\ \text{II} &: y_i = 0.5 + (\boldsymbol{\beta}^\top \boldsymbol{x}_i + 0.8)\epsilon_i, \\ \text{III} &: y_i = 0.5 + 0.5(\boldsymbol{\beta}^\top \boldsymbol{x}_i)^3 + 0.8\epsilon_i, \end{aligned}$$

where $\epsilon_i \sim N(0, 1)$. For the contaminated sample, i.e., $i = n_0 + 1, \ldots, n$, we generate the response based on the same models but the covariate and noise are based on the contamination type:

- **leverage only:** set $\boldsymbol{x}_i \sim t_{10}(5, \boldsymbol{\Sigma})$ and $\epsilon_i \sim N(0, 1)$.
- **response outlier only:** set $\boldsymbol{x}_i \sim N(\mathbf{0}, \boldsymbol{\Sigma})$ and $\epsilon_i \sim \chi^2_5$.
- **both leverage and response:** set $\boldsymbol{x}_i \sim t_{10}(5, \boldsymbol{\Sigma})$ and $\epsilon_i \sim \chi^2_5$.

Both models I and III are homoscedastic with monotone response links, thus, we expect the penalized OLS and SIR to perform very well. Model II is heteroscedastic with a constant mean. Thus, we expect both OLS and SIR to struggle to recover the direction. However, SOPG-LAD does not suffer this limitation. For all models and settings, we evaluate the performance of each method based on the average size of the active set

$$\hat{K} = \left|\{j \in \{1, \ldots, p\} : \hat{b}_j \neq 0\}\right|$$

and the false positive rate (FPR) given by

$$\text{FPR} = \frac{\left|\{j \in \{1, \ldots, p\} : \hat{b}_j \neq 0 \wedge \beta_j = 0\}\right|}{\left|\{j \in \{1, \ldots, p\} : \beta_j = 0\}\right|} \times 100.$$

We implement OLS-LASSO, OLS-SCAD, OLS-MCP using the *ncvreg* package while the SIR-LASSO is implemented using the *LassoSIR* package. The tuning parameters are based on cross-validation with the optimal λ being the one the achieves the minimum MSE, which are readily available in the packages. For SIR-LASSO, we fix the number of slices to 10 for

all settings. The sparse local median regression is implemented using the *conquer*. While the *conquer* package offers readily available cross-validation functions, it is very slow. Thus, we fix the $\lambda = 0.01$ for $p = 10$ and $\lambda = 0.1$ for large $p = \{200, 1000\}$ based our preliminary simulation. Tables 1–3 summarize the simulation results.

Table 1
Variable selection performance under Model I. Results for the best method are emboldened.

			Leverage only				Response outliers only				Leverage & response outliers			
			p=10		p=200		p=10		p=200		p=10		p=200	
n	*pout*	Method	$\hat{K}$	FPR	$\hat{K}$	FPR	$\hat{K}$	FPR	$\hat{K}$	FPR	$\hat{K}$	FPR	$\hat{K}$	FPR
50	0.02	OLS-LASSO	8.1	61.60	19.2	7.35	8.1	61.60	19.2	7.4	8.1	61.60	19.2	7.35
		OLS-SCAD	5.8	15.80	9.3	2.24	5.8	15.80	9.3	2.2	5.8	15.80	9.3	2.24
		OLS-MCP	5.7	14.60	6.5	0.91	5.7	14.60	6.5	0.9	5.7	14.60	6.5	0.91
		SIR-LASSO	7.6	60.20	19.5	8.63	7.5	59.80	19.5	8.6	7.5	59.80	19.5	8.63
		SOPG-LAD	**4.9**	**6.20**	**4.6**	**0.22**	**4.9**	**6.20**	**4.6**	**0.2**	**4.9**	**6.20**	**4.6**	**0.22**
	0.05	OLS-LASSO	8.0	59.60	13.8	4.59	7.6	53.80	21.6	8.6	8.2	65.60	14.90	5.20
		OLS-SCAD	5.9	17.60	**5.5**	**0.37**	5.5	15.80	10.7	3.2	6.0	24.80	6.60	1.01
		OLS-MCP	5.8	17.20	**5.3**	**0.39**	5.3	12.20	6.6	1.2	6.1	25.40	**5.50**	**0.62**
		SIR-LASSO	5.7	43.60	8.2	3.54	7.2	58.40	18.8	8.5	5.2	41.80	8.90	3.99
		SOPG-LAD	**4.8**	**5.00**	3.8	0.41	**5.1**	**8.40**	**4.6**	**0.3**	**4.8**	**7.00**	3.80	0.49
			p=10		p=200		p=10		p=200		p=10		p=200	
n	*pout*	Method	$\hat{K}$	FPR	$\hat{K}$	FPR	$\hat{K}$	FPR	$\hat{K}$	FPR	$\hat{K}$	FPR	$\hat{K}$	FPR
100	0.02	OLS-LASSO	8.1	62.40	22.4	1.75	8.2	63.60	31.6	2.67	8.3	65.80	25.90	2.10
		OLS-SCAD	5.5	10.80	7.2	0.22	5.6	12.20	14.6	0.96	5.8	15.60	14.30	0.90
		OLS-MCP	5.6	11.40	5.9	0.09	5.7	14.20	7.6	0.27	5.9	18.20	7.90	0.30
		SIR-LASSO	8.0	61.60	12.8	1.12	8.1	64.20	33.1	2.98	7.9	60.40	13.60	1.18
		SOPG-LAD	**4.8**	**1.40**	**4.6**	**0.01**	**4.7**	**1.80**	**4.7**	**0.01**	**5**	**3.00**	**4.60**	**0.01**
	0.05	OLS-LASSO	8.2	64.60	14.1	0.92	8.1	61.80	30.8	2.60	8.4	67.80	20.40	1.55
		OLS-SCAD	5.5	10.40	**5.5**	**0.05**	5.8	18.20	16	1.11	6	19.80	9.20	0.42
		OLS-MCP	5.8	15.20	**5.3**	**0.03**	5.7	16.40	8.1	0.32	6.2	23.60	5.90	0.10
		SIR-LASSO	7.3	54.80	10.3	0.89	8.4	69.00	35.2	3.21	7.5	59.40	14.40	1.32
		SOPG-LAD	**4.9**	**1.80**	**4.4**	**0.02**	**4.9**	**2.60**	**4.6**	**0.01**	**5**	**4.20**	**4.30**	**0.03**

Since the true active set size is fixed at $K = 5$, effective recovery is characterized by $\hat{K}$ close to 5 and a low false positive rate (FPR). Under Model I, which corresponds to a correctly specified linear mean structure, SOPG-LAD consistently delivers more parsimonious models and markedly lower false positive rates than OLS-LASSO, OLS-SCAD, and OLS-MCP, as evidenced in Table 1. In low dimensions, OLS-LASSO systematically over-selects variables, while OLS-SCAD and OLS-MCP only partially mitigate this effect, particularly as the contamination proportion increases. SIR-LASSO shows similar results to OLS-SCAD and OLS-MCP, and sometimes yield worse estimates comparatively. In contrast, SOPG-LAD maintains estimated model sizes close to the true value and achieves uniformly smaller FPRs across all contamination mechanisms. These advantages become more pronounced in the high-dimensional setting, where SOPG-LAD exhibits near-zero FPRs for $n = 100$, whereas penalized OLS and SIR methods continue to include substantial number of irrelevant variables.

Model II presents more challenge for OLS and SIR as they struggle to recover directions in the variance component. Therefore, it is not surprising that the penalized OLS methods perform woefully as shown in Table 2. In low-dimensions $p = 10$, both SIR-LASSO and SOPG-LAD appear to select only two active covariates on average. In high-dimensions, SOPG-LAD stabilizes with estimates very close to the oracle. SIR-LASSO on the other hand, selects about three times the number of active participants, indicating a very high level of instability.

Table 2
Variable selection performance under Model II

			Leverage only				Response outliers only				Leverage & response outliers			
			p=10		p=1000		p=10		p=1000		p=10		p=1000	
n	*pout*	Method	$\hat{K}$	FPR	$\hat{K}$	FPR	$\hat{K}$	FPR	$\hat{K}$	FPR	$\hat{K}$	FPR	$\hat{K}$	FPR
50	0.02	OLS-LASSO	0.9	8.40	3.1	1.51	0.9	8.40	3.1	1.51	0.9	8.40	3.1	1.51
		OLS-SCAD	0.9	7.80	2.0	0.96	0.9	7.80	2.0	0.96	0.9	7.80	2.0	0.96
		OLS-MCP	0.5	3.80	1.1	0.53	0.5	3.80	1.1	0.53	0.5	3.80	1.1	0.53
		SIR-LASSO	**4.5**	**44.00**	17.5	8.75	**4.5**	**44.00**	17.5	8.75	**4.5**	**44.00**	17.5	8.75
		SOPG-LAD	**5.4**	**50.20**	**6.0**	**3.00**	**5.4**	**50.20**	**6.0**	**3.00**	**5.4**	**50.20**	**6.0**	**3.00**
	0.05	OLS-LASSO	1.4	15.40	2.4	1.14	1.1	9.40	3.0	1.44	**4.5**	**42.00**	14.2	7.04
		OLS-SCAD	0.9	9.20	1.8	0.88	0.7	6.60	2.2	1.05	3.4	33.40	9.2	4.62
		OLS-MCP	0.8	8.80	0.6	0.25	0.8	7.80	1.0	0.48	2.8	28.00	3.4	1.68
		SIR-LASSO	**3.3**	**34.20**	**6.4**	**3.16**	**4.2**	**43.00**	17.7	8.87	**4.8**	**47.60**	10.0	4.96
		SOPG-LAD	**5.2**	**52.60**	3.8	1.88	**5.4**	**50.00**	**6.3**	**3.15**	**5.1**	**49.80**	**6.7**	**3.34**
			p=10		p=1000		p=10		p=1000		p=10		p=1000	
n	*pout*	Method	$\hat{K}$	FPR	$\hat{K}$	FPR	$\hat{K}$	FPR	$\hat{K}$	FPR	$\hat{K}$	FPR	$\hat{K}$	FPR
100	0.02	OLS-LASSO	1.2	11.60	**5.5**	**0.55**	0.8	8.60	**4.8**	**0.47**	**4.2**	**40.00**	28.7	2.86
		OLS-SCAD	0.6	5.40	3.1	0.31	0.8	7.20	3.2	0.32	3.6	36.20	19.0	1.90
		OLS-MCP	0.7	6.20	1.1	0.11	0.7	6.80	1.2	0.12	2.9	30.40	**6.8**	**0.67**
		SIR-LASSO	**3.8**	**35.60**	11.4	1.13	**4.1**	**41.60**	37.6	3.76	4.4	43.60	14.0	1.39
		SOPG-LAD	**5.1**	**51.20**	**4.8**	**0.48**	**5.3**	**51.00**	7.0	0.69	**4.9**	**47.40**	7.6	0.76
	0.05	OLS-LASSO	1.3	12.60	**6.4**	**0.64**	1.2	9.80	**5.6**	**0.55**	5.8	56.20	30.7	3.05
		OLS-SCAD	0.9	10.00	2.4	0.24	1.0	7.60	4.0	0.39	4.3	42.40	20.0	2.00
		OLS-MCP	0.9	9.00	0.3	0.03	1.0	7.60	1.4	0.13	3.6	36.40	**5.1**	**0.50**
		SIR-LASSO	**3.5**	**33.60**	8.7	0.86	4.2	41.00	38.0	3.80	4.9	46.40	16.2	1.61
		SOPG-LAD	**4.7**	**41.40**	3.5	0.35	5.3	47.00	6.8	0.68	4.6	46.20	8.3	0.83

Table 3
Variable selection performance under Model III

			Leverage only				Response outliers only				Leverage & response outliers			
			p=10		p=1000		p=10		p=1000		p=10		p=1000	
n	*pout*	Method	$\hat{K}$	FPR	$\hat{K}$	FPR	$\hat{K}$	FPR	$\hat{K}$	FPR	$\hat{K}$	FPR	$\hat{K}$	FPR
50	0.02	OLS-LASSO	8.0	66.40	19.1	7.90	8.0	66.40	19.1	7.90	8.0	66.40	19.1	7.90
		OLS-SCAD	6.3	41.40	11.2	4.00	6.3	41.40	11.2	4.00	6.3	41.40	11.2	4.00
		OLS-MCP	5.9	36.20	6.7	2.10	5.9	36.20	6.7	2.10	5.9	36.20	6.7	2.10
		SIR-LASSO	7.8	64	20.7	9.21	7.8	64.00	20.7	9.21	7.8	64.00	20.7	9.21
		SOPG-LAD	**3.6**	**7.80**	**4.7**	**0.60**	**3.6**	**7.80**	**4.7**	**0.60**	**3.6**	**7.80**	**4.7**	**0.60**
	0.05	OLS-LASSO	6.4	53.80	16.1	6.90	7.8	62.60	20.4	8.70	6.3	53.60	16.0	6.90
		OLS-SCAD	4.3	30.80	6.0	2.20	6.0	34.00	12.3	4.70	4.8	36.40	6.4	2.40
		OLS-MCP	4.6	34.20	3.7	1.20	5.7	29.40	6.3	1.90	4.2	28.40	3.6	1.20
		SIR-LASSO	6.0	43.80	8.0	3.43	7.7	62.40	18.7	8.31	6.2	47.20	7.9	3.37
		SOPG-LAD	**2.6**	**8.60**	**5.4**	**1.50**	**3.6**	**8.80**	**4.7**	**0.60**	**2.6**	**8.40**	**5.3**	**1.50**
			p=10		p=1000		p=10		p=1000		p=10		p=1000	
n	*pout*	Method	$\hat{K}$	FPR	$\hat{K}$	FPR	$\hat{K}$	FPR	$\hat{K}$	FPR	$\hat{K}$	FPR	$\hat{K}$	FPR
100	0.02	OLS-LASSO	8.2	70.40	30.3	2.70	8.0	61.40	33.4	2.90	8.0	68.60	31.2	2.80
		OLS-SCAD	6.1	39.00	18.8	1.60	6.2	31.20	25.4	2.10	6.0	38.00	19.0	1.60
		OLS-MCP	5.9	36.20	6.9	0.40	5.8	25.60	10.6	0.70	5.9	35.60	7.4	0.50
		SIR-LASSO	7.8	58.6	12.9	1.118	8.4	68.00	35.5	3.21	7.8	57.40	13.2	1.14
		SOPG-LAD	**3.2**	**5.20**	**4.5**	**0.10**	**3.4**	**4.00**	**4.6**	**0.00**	**3.2**	**5.20**	**4.5**	**0.10**
	0.05	OLS-LASSO	7.3	60.60	29.6	2.70	8.1	64.80	31.8	2.80	7.3	60.80	29.6	2.70
		OLS-SCAD	5.4	36.60	11.8	1.00	6.2	30.60	26.2	2.20	5.3	34.00	11.3	0.90
		OLS-MCP	5.3	35.40	4.1	0.20	6.0	28.80	9.8	0.60	5.4	36.80	4.3	0.20
		SIR-LASSO	7.8	62.20	10.0	0.86	8.3	66.60	34.7	3.14	7.8	61.00	10.3	0.89
		SOPG-LAD	**2.6**	**5.20**	**4.6**	**0.20**	**3.3**	**3.00**	**4.6**	**0.00**	**2.5**	**4.80**	**4.6**	**0.20**

Model III is non-linear but still monotone. Thus, as $\mathbf{X}$ satisfies the linear conditional mean assumption, OLS should be still be able to recover the direction in the mean component

despite the model misspecification [19]. Because of the cubic function in the mean, leverage points are expected to have exaggerated effect on the estimates. Again, we see in Table 3 that SOPG-LAD outperforms the penalized OLS and SIR methods, especially in high dimensions.

Overall, the results in Tables 1–3 indicate that SOPG-LAD provides a more reliable balance between sparsity and robustness than existing penalized OLS and SIR methods. Its advantages are most evident in high-dimensional settings and under complex contamination structures, where conventional approaches either over-select spurious variables or fail to recover the true signal. These findings support the use of SOPG-LAD as a robust variable–selection tool in contaminated and high-dimensional regression problems.

6. Real Application

In this section, we consider two real applications. For each application, we employ variable–selection techniques to determine the most important variables. We then fit a classical OLS on the selected variables and evaluate prediction based on the resulting smaller model. Each dataset is repeatedly partitioned into training and test sets to allow for the evaluation of out-of-sample performance.

We utilize partial distance correlation [29] to quantify the conditional independence in (4) since we do not know the true active set. For a random sample (s, t, w) from a joint distribution of (S, T, W), [29] defined the sample partial distance correlation of S and T given W as

$$R^*(s,t,w) = \begin{cases} \dfrac{(P_{w^\perp}(s) \cdot P_{w^\perp}(t))}{\|P_{w^\perp}(s)\|\|P_{w^\perp}(t)\|}, & \|P_{w^\perp}(s)\|\|P_{w^\perp}(t)\| \neq 0, \\ 0, \text{ otherwise}, \end{cases} \tag{13}$$

where $P_{w^\perp}(s)$ and $P_{w^\perp}(t)$ belong to the Hilbert space $\mathcal{H}_n$. The partial distance correlation can accommodate variables of arbitrary dimensions, which makes it suitable for our case. Here, we are interested in $R^*(Y, \mathbf{X}^\top_{\hat{\mathcal{A}}^c}, \mathbf{X}^\top_{\hat{\mathcal{A}}})$, where $\hat{\mathcal{A}}$ denote the estimated active set. If $\mathbf{X}^\top_{\hat{\mathcal{A}}}$ captures all the relevant covariates, then Y should not depend on $\mathbf{X}^\top_{\hat{\mathcal{A}}^c}$, which will result in a zero partial distance correlation. Thus, smaller estimates of the partial distance correlation indicate better performance.

We complement the partial distance correlation measure with a root mean squared prediction error to quantify the out-of-sample performance. Let $\hat{y}_i$ denote the predicted response for the ith observation. The root mean squared prediction error (RMSPE) is given by

$$\text{RMSPE} = \left\{ \frac{1}{n_{\text{te}}} \sum_{i \in \mathcal{I}_{\text{te}}} \left(y_i - \hat{y}_i\right)^2 \right\}^{1/2}, \tag{14}$$

where $\mathcal{I}_{\text{te}}$ is the set of indices representing the test set and n_{te} is the size of the test set.

6.1. Application to Pollution Data

We analyze the US air–pollution dataset previously studied by [11] and [35]. The data comprise $n = 60$ cities (year 1960) with response y as the total age–adjusted mortality rate (per 100,000) and $p = 15$ covariates summarizing meteorological, demographic, socioeconomic,

and pollution-related factors. Standard collinearity diagnostics indicate substantial linear dependence among covariates: variance inflation factors (VIFs) are extremely large for the hydrocarbon (HC) and nitric oxide (NOX) pollution potential variables (approximately 98.6 and 105.0, respectively), consistent with near-redundant information in those measures. Influence diagnostics further suggest that a small number of observations may drive least-squares fitting: using the usual leverage cutoff $2(p+1)/n = 0.533$, several cities exhibit high leverage, and Cook's distance highlights cities 7, 29, and 32 as particularly influential (Figure 1). Taken together, these features motivate estimation procedures that (i) remain stable under severe multicollinearity and (ii) can perform variable selection without relying on a single potentially influential subset of observations.

The data is randomly partitioned repeatedly into 40 training observations and a test sample of 20. All tuning and model fitting are performed *exclusively* on the training set. covariates are also centered and scaled before model fitting. In addition to the methods already compared in the simulation study, we include the classical OLS as $p < n$. For the classical OLS, estimates with p-values below 0.05 are taken to be significant (active). The post-selection RMSPE is based on classical OLS on the active covariates. The number of slices for SIR-LASSO is fixed at 5 and for SOPG-LAD, we fixed $\lambda = 0.1$ and the tolerance level at 0.005.

Table 4

Train selection proportion in 100 random samples with average root mean squared prediction error and partial distance correlation in test sample. Proportions that are at least 50% are emboldened.

covariate	**OLS**	**OLS-LASSO**	**OLS-SCAD**	**OLS-MCP**	**SIR-LASSO**	**SOPG-LAD**
Annual precipitation	**0.67**	**0.97**	**0.61**	**0.67**	**0.81**	**0.86**
Jan. Temp	**0.66**	**0.74**	**0.74**	**0.67**	**0.58**	0.28
Jul. Temp	**0.61**	**0.52**	0.49	0.44	0.28	0.02
% pop. aged 65+	**0.54**	0.11	0.22	0.18	0.13	0.29
Household size	**0.73**	0.23	0.29	0.29	**0.58**	**0.60**
Education	**0.70**	**0.86**	**0.64**	**0.62**	**0.73**	0.21
% housing units	**0.59**	**0.55**	0.18	0.13	0.45	0.46
Urban pop.	0.37	**0.82**	**0.55**	**0.50**	**0.71**	**0.98**
% NW urban pop.	0.25	**1.00**	**0.99**	**1.00**	**1.00**	0.29
% White collar	0.17	**0.52**	0.29	0.26	0.48	0.14
% Income <$30K	0.07	0.19	0.12	0.08	0.20	**0.98**
Rel. HC pollution	0.01	0.13	0.08	0.09	0.24	0.01
Rel. NOX pollution	**0.67**	0.11	0.07	0.08	0.07	**0.86**
Rel. SOX pollution	**0.66**	**0.98**	**0.86**	**0.78**	**0.95**	0.28
% Rel. humidity 1pm	**0.61**	0.34	0.16	0.22	0.29	0.02
Avg. RMSPE	53.257	45.939	45.897	46.84	45.355	**43.566**
Avg. $R^*(Y, \mathbf{X}^\top_{\hat{\mathcal{A}}^c}, \mathbf{X}^\top_{\hat{\mathcal{A}}})$	0.087	0.081	0.078	0.089	0.085	**0.066**

Table 4 summarizes selection proportions alongside the average test RMSPE and average partial distance correlation of Y and $\mathbf{X}^\top_{\hat{\mathcal{A}}^c}$ given $\mathbf{X}^\top_{\hat{\mathcal{A}}}$. Several patterns emerge. We see that OLS selection proportions are high across many covariates but with notably larger average test RMSPE and partial distance correlation compared to the penalized competitors. This is plausibly due to the instability induced by the severe multicollinearity and the presence of influential observations, as further illustrated by the residual diagnostics in Figure 1. OLS-LASSO appears to be very similar to the OLS, which is also not surprising as OLS-LASSO is known for over-selection in correlated designs. Among the remaining approaches, sparsity-inducing methods generally concentrate selection on a smaller subset of covariates. In particular, SIR-LASSO exhibits more focused selection behavior than OLS-based penalization, reflecting the

benefits of incorporating sufficient dimension reduction prior to regularization. The method tends to retain variables associated with urbanization and pollution intensity, suggesting that the dimension-reduction step effectively captures key directions of variation in the response. Nevertheless, SIR-LASSO remains sensitive to influential observations, as evidenced by its comparatively higher average test RMSPE and partial distance correlation relative to SOPG-LAD.

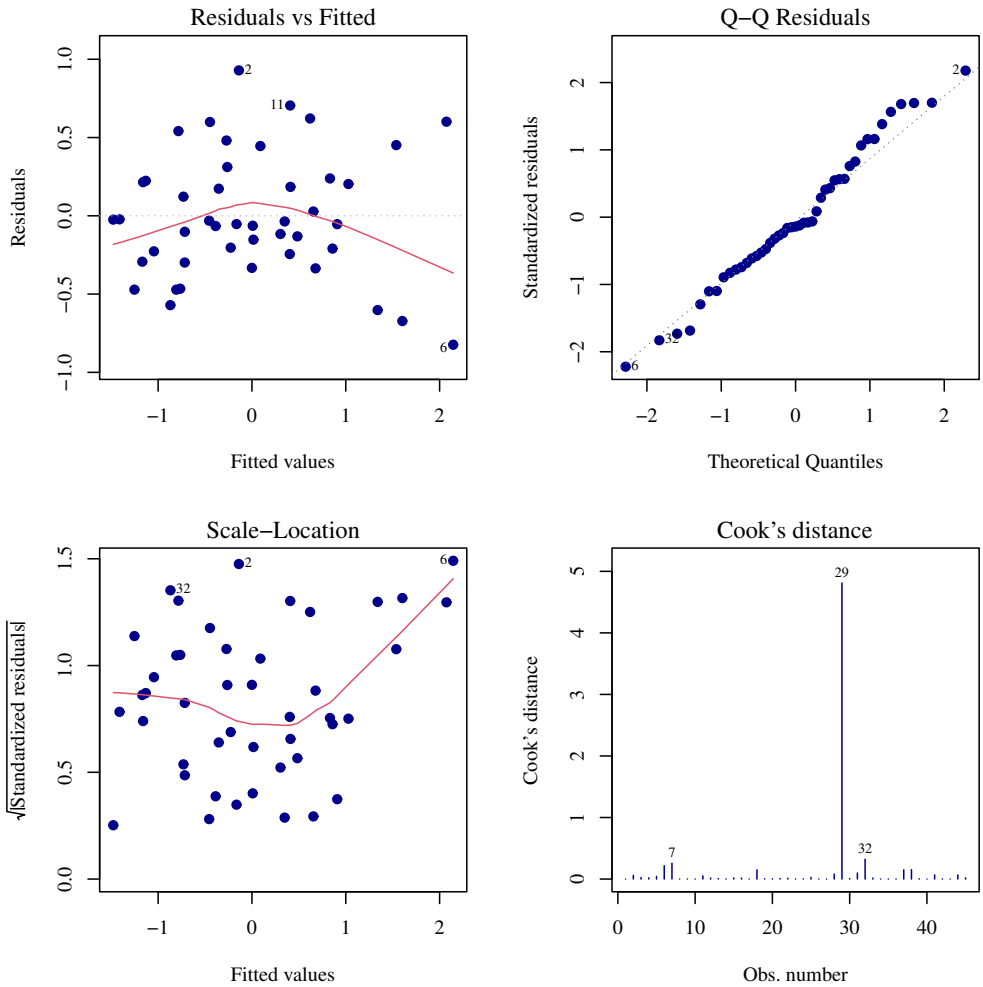

Fig 1. *Residual diagnostics for the pollution training data*

Across all methods, socio-demographic variables related to urban structure (e.g., percentage of urban population) are frequently selected, whereas several meteorological variables are downweighted or excluded. The proposed SOPG-LAD method achieves the strongest overall performance, attaining the smallest average test RMSPE (43.566) and the lowest average partial distance correlation (0.066). Taken together, these results indicate that SOPG-LAD delivers improved predictive accuracy while yielding a comparatively sparse and stable selection pattern in a setting characterized by strong linear dependencies (notably for Nitric Oxide) and a small number of influential observations. Overall, the empirical evidence supports

SOPG-LAD as a robust alternative to both classical penalization techniques and SDR-based sparse methods such as SIR-LASSO when stable prediction under severe contamination is the primary objective.

6.2. Application to Riboflavin Data

We next consider the *riboflavin* dataset, a high-dimensional genomic study on *Bacillus subtilis* that has become a standard benchmark for sparse regression in the $p \gg n$ regime. The data were introduced in [5] and are distributed in `R` via the `hdi` package. The goal is to identify genes whose expression levels are associated with increased riboflavin (*vitamin B2*) production, with the longer-term objective of engineering higher-yield strains. The sample size is $n = 71$ and the number of covariates is $p = 4088$, where each covariate is the (log-)expression level of a gene. The response is the log riboflavin production rate.

Because p is orders of magnitude larger than n, we evaluate only the penalized methods under repeated random splitting. For each of 100 Monte Carlo replicates, we draw a training set of size 50 and use the remaining 21 observations for testing. covariates are standardized before model fitting and the number of slices for SIR-LASSO is fixed at 10. For SOPG-LAD, we maintain $\lambda = 0.1$ and a tolerance level of 0.005.

Table 5

Riboflavin data ($n = 71$, $p = 4088$): top 10 selected genes (100 splits) with average test RMSPE and partial distance correlation. Genes selected by at least 3 methods are highlighted in red while those highlighted in blue are selected by two methods. Genes in black color were selected by only one method.

OLS-LASSO		OLS-SCAD		OLS-MCP		SIR-LASSO		SOPG-LAD	
Gene	Freq	Gene	Freq	Gene	Freq	Gene	Freq	Gene	Freq
1	1.00	**1**	0.72	**1**	0.78	**932**	0.51	**932**	0.89
3977	0.83	**2126**	0.42	**214**	0.47	**295**	0.48	**2273**	0.86
2126	0.8	**3568**	0.33	2270	0.3	**2273**	0.43	**214**	0.66
3568	0.79	**214**	0.31	3959	0.16	3469	0.37	**295**	0.59
3229	0.72	**2272**	0.30	69	0.13	**1439**	0.34	2543	0.49
2272	0.67	241	0.2	668	0.13	1582	0.33	**660**	0.46
230	0.61	2	0.19	**660**	0.11	**3568**	0.33	**3568**	0.36
963	0.46	1067	0.19	579	0.10	**214**	0.31	816	0.33
1439	0.46	**3977**	0.19	3917	0.10	1845	0.30	3618	0.32
1920	0.46	**660**	0.18	1696	0.08	**3656**	0.29	**3656**	0.30
Avg. RMSPE	**14.041**		14.050		14.058		14.050		14.061
Avg. $R^*(Y, \mathbf{X}_{\hat{\mathcal{A}}^c}^\top, \mathbf{X}_{\hat{\mathcal{A}}}^\top)$	0.091		0.087		0.102		0.087		**0.072**

Table 5 reports the ten most frequently selected genes for each method, together with average test RMSPE and average partial distance correlation. Several features merit attention. The five procedures produce partially overlapping, but not identical sets of frequently selected genes. Some genes appear consistently under multiple penalties, suggesting robust signals that persist across different regularization geometries. The average test RMSPE is very similar across the methods (approximately 14.04–14.06 in the reported runs), indicating that, for this dataset, the different selection mechanisms lead to comparable predictive accuracy. This pattern is typical in $p \gg n$ gene-expression problems, where multiple sparse representations can achieve near-equivalent prediction due to strong covariate dependence and the presence of many correlated gene groups.

However, the partial distance correlation favors SOPG-LAD in our experiments (average ≈ 0.072), suggesting that the variables retained by SOPG-LAD yield predictions that are more conditionally aligned with the response after accounting for the remaining covariates. In other words, SOPG-LAD appears to identify a subset whose retained signal is less "replaceable" by the excluded genes, which is a desirable feature when interpretability and mechanistic follow-up are priorities. By contrast, MCP yields the largest average partial distance correlation in these runs, which can occur when the selected subset leaves residual dependence structure that is still captured by excluded but correlated genes.

Taken together, these results suggest that the main differences among the methods for the riboflavin data are not driven by predictive accuracy (which is essentially tied across the procedures), but by *which genes* are repeatedly retained and how conditionally informative the selected subset appears to be. This distinction matters in genomics applications where the downstream task is not only prediction but also prioritization of candidate genes for experimental validation. Within this framing, SOPG-LAD offers a competitive alternative as it achieves prediction comparable to popular convex and nonconvex penalties while producing a selected set that, in our experiments, attains improved conditional dependence diagnostics.

7. Conclusion

This paper investigates the impact of influential observations and leverage effects on variable selection in single-index models, particularly in settings characterized by strong dependence among covariates and the use of modern shrinkage penalties. Our findings indicate that commonly used penalized least-squares (LASSO, MCP, SCAD) and sliced inverse regression methods are vulnerable to contamination, often leading to unstable model selection and inflated false discoveries. These difficulties are especially pronounced in heteroscedastic and high-dimensional regimes, where standard procedures struggle to distinguish relevant predictors from spurious signals.

To address these limitations, we introduced the sparse outer product of gradients based on locally weighted median regression (SOPG-LAD). The proposed method combines robust local gradient estimation with sparse low-rank aggregation, thereby incorporating case-influence information directly into the variable selection process. By enforcing sparsity at both the local and global levels, SOPG-LAD yields a selection mechanism that remains stable in the presence of leverage points, heavy-tailed errors, and strong predictor correlations.

Simulation studies across a broad range of contamination patterns, dimensional settings, and model complexities demonstrate that SOPG-LAD provides more reliable sparsity recovery than existing penalized approaches based on ordinary least squares and sliced inverse regression. In particular, the method maintains accurate model size and low false positive rates in regimes where competing procedures become increasingly unstable. These advantages are further supported by real data analyses. In the air pollution study, SOPG-LAD delivers consistent selection and strong predictive performance in the presence of influential observations and multicollinearity. In the riboflavin dataset, the method achieves competitive prediction accuracy while retaining variables that exhibit stronger conditional associations in a high-dimensional genomic context.

Taken together, these results suggest that explicitly incorporating influence diagnostics into gradient-based dimension reduction provides a practical and effective route to robust

variable selection in single-index models. SOPG-LAD offers a complementary alternative to conventional penalization strategies when stability under contamination is a primary concern.

The proposed framework is currently limited to single-index structures. Extending the methodology to multi-index models, as well as to settings involving grouped or hierarchical signals, represents an important direction for future research. Further work may also explore adaptive weighting schemes and alternative robust loss functions to enhance performance under more complex forms of data contamination.

Appendix of Proofs

Proof of Proposition 3.1. See Lemma 1 in [15].

□

Proof of Proposition 3.2.

$$\frac{\partial Q_\tau(Y \mid \boldsymbol{x})}{\partial \boldsymbol{x}} = \frac{\partial Q_\tau(Y \mid \boldsymbol{\beta}^\top \boldsymbol{x})}{\partial \boldsymbol{x}} = \frac{\partial Q_\tau(Y \mid \boldsymbol{\beta}^\top \boldsymbol{x})}{\partial(\boldsymbol{\beta}^\top \boldsymbol{x})}\frac{\partial \boldsymbol{\beta}^\top \boldsymbol{x}}{\partial \boldsymbol{x}} = \boldsymbol{\beta}\frac{\partial Q_\tau(Y \mid \boldsymbol{\beta}^\top \boldsymbol{x})}{\partial(\boldsymbol{\beta}^\top \boldsymbol{x})},$$

which is a vector in $\text{span}(\boldsymbol{\beta}) = \mathcal{S}_{Y|\mathbf{X}}$.

□

Proof of Lemma 4.1.
The proof follows directly by the convexity Lemma [23], which establishes that

$$\sup_{\boldsymbol{\theta}\in\boldsymbol{\Theta}} |\mathcal{L}_n(\boldsymbol{\theta}, \mathbf{Z}; \boldsymbol{x}_0) - \mathcal{L}(\boldsymbol{\theta}, \mathbf{Z}; \boldsymbol{x}_0)| \xrightarrow{p} 0$$

for each $\boldsymbol{\theta}$ at any given $\mathbf{X} = \boldsymbol{x}_0$, where $\mathcal{L}_n(.)$ is the sample version of $\mathcal{L}$.

□

Proof of Theorem 4.2.
Notice that

$$\|\tilde{\mathbf{B}}_n - \boldsymbol{u}_1\boldsymbol{\eta}\|_F^2 = \text{tr}(\tilde{\mathbf{B}}_n\tilde{\mathbf{B}}_n^\top) - 2\boldsymbol{\eta}^\top\tilde{\mathbf{B}}_n^\top\boldsymbol{u}_1 + \|\boldsymbol{u}_1\|^2\|\boldsymbol{\eta}\|^2,$$

which implies the minimization in (10) is equivalent to minimizing

$$-2\boldsymbol{\eta}^\top\tilde{\mathbf{B}}_n^\top\boldsymbol{u}_1 + \|\boldsymbol{u}_1\|^2\|\boldsymbol{\eta}\|^2 + P_\xi(\boldsymbol{\eta})$$

We also see that $\boldsymbol{\eta}$ depends on $\tilde{\mathbf{B}}_n$ only through $\tilde{\mathbf{B}}_n^\top\tilde{\mathbf{B}}_n$, see [25].

Next, for any fixed $\boldsymbol{u}_1$, let

$$W = \tilde{\mathbf{B}}_n^\top\boldsymbol{u}_1 = \boldsymbol{\eta} + \epsilon, \text{ where }, \epsilon \sim N\left(\mathbf{0}, \frac{\sigma^2}{n}\mathbf{I}_p\right),$$

which is a standard intercept only penalized least squares. If we choose $\xi \asymp \frac{\sigma}{\sqrt{n}}\sqrt{\log(p/s)}$, we obtain the optimal rate of $\frac{s\log(p/s)}{n}$ for $\hat{\boldsymbol{\eta}}$ when $P_\xi(\boldsymbol{\eta}) = \xi\|\boldsymbol{\eta}\|_1$ [1]. Similarly, if optimal rates are chosen for different penalties such as SCAD, we expect to obtain rates similar or better. Thus, as $\boldsymbol{\Lambda}_n$ is consistent for $\boldsymbol{\Lambda}$, $\text{span}(\hat{\boldsymbol{\eta}}) \subseteq \mathcal{S}_{Y|\mathbf{X}}$, which leads to the desired results. □

Supplementary Material

Code for reproducing results

The code for the SOPG-LAD is publicly available at https://github.com/ansoale/SOPG-LAD.

References

[1] BELLEC, P. C., LECUÉ, G. and TSYBAKOV, A. B. (2018). Slope meets Lasso: Improved oracle bounds and optimality. *The Annals of Statistics* **46** 3603 – 3642. https://doi.org/10.1214/17-AOS1670

[2] BELLONI, A. and CHERNOZHUKOV, V. (2011). ℓ_1-penalized quantile regression in high-dimensional sparse models. *The Annals of Statistics* **39**.

[3] BICKEL, P. J., RITOV, Y. and TSYBAKOV, A. B. (2009). Simultaneous analysis of Lasso and Dantzig selector. *The Annals of Statistics* **37** 1705 – 1732. https://doi.org/10.1214/08-AOS620

[4] BINDELE, H. F., ABEBE, A. and ZENG, P. (2019). Robust estimation and selection for single-index regression model. *Journal of Statistical Computation and Simulation* **89** 1376–1393.

[5] BÜHLMANN, P. (2014). High-dimensional statistics with a view toward applications in biology. *Annual Review of Statistics and Its Application* **1** 255–278.

[6] COOK, R. D. (1977). Detection of Influential Observation in Linear Regression. *Technometrics* **19** 15–18.

[7] FAN, J. and GIJBELS, I. (1996). *Local polynomial modelling and its applications*. Chapman & Hall/CRC.

[8] FAN, J. and LI, R. (2001). Variable Selection via Nonconcave Penalized Likelihood and Its Oracle Properties. *Journal of the American Statistical Association* **96** 1348–1360.

[9] FAN, Y. and GUERRE, E. (2016). Multivariate local polynomial estimators: Uniform boundary properties and asymptotic linear representation. In *Essays in Honor of Aman Ullah*, **36** 489–537. Emerald Group Publishing Limited.

[10] FENG, L., ZOU, C. and WANG, Z. (2012). Rank-based inference for the single-index model. *Statistics & Probability Letters* **82** 535–541.

[11] GENÇ, M. and LUKMAN, A. (2025). Weighted LAD-Liu-LASSO for robust estimation and sparsity. *Computational Statistics* 1–30.

[12] ICHIMURA, H. (1993). Semiparametric least squares (SLS) and weighted SLS estimation of single-index models. *Journal of econometrics* **58** 71–120.

[13] KIM, H., WU, Y. and SHIN, S. J. (2019). Quantile-slicing estimation for dimension reduction in regression. *Journal of Statistical Planning and Inference* **198** 1–12.

[14] KONG, E. and XIA, Y. (2007). Variable selection for the single-index model. *Biometrika* **94** 217–229.

[15] KONG, E. and XIA, Y. (2014). An adaptive composite quantile approach to dimension reduction. *Ann. Stat.* **42** 1657–1688.

[16] LI, B. (2018). *Sufficient dimension reduction: Methods and applications with R*. CRC Press.

[17] LI, B., ARTEMIOU, A. and LI, L. (2011). Principal support vector machines for linear and nonlinear sufficient dimension reduction. *The Annals of Statistics* **39** 3182 – 3210. https://doi.org/10.1214/11-AOS932

[18] LI, K.-C. (1991). Sliced inverse regression for dimension reduction. *Journal of the American Statistical Association* **86** 316–327.

[19] LI, K.-C. and DUAN, N. (1989). Regression analysis under link violation. *The Annals of Statistics* **17** 1009–1052.

[20] LIN, Q., ZHAO, Z. and LIU, J. S. (2019). Sparse sliced inverse regression via lasso. *Journal of the American Statistical Association* **114** 1726–1739.

[21] MAMMEN, E., VAN KEILEGOM, I. and YU, K. (2019). Expansion for moments of regression quantiles with applications to nonparametric testing. *Bernoulli* **25** 793–827.

[22] PEÑA, D. (2005). A New Statistic for Influence in Linear Regression. *Technometrics* **47** 1–12.

[23] POLLARD, D. (1991). Asymptotics for least absolute deviation regression estimators. *Econometric Theory* **7** 186–199.

[24] PRENDERGAST, L. A. (2008). Trimming influential observations for improved single-index model estimated sufficient summary plots. *Computational statistics & data analysis* **52** 5319–5327.

[25] SHEN, H. and HUANG, J. Z. (2008). Sparse principal component analysis via regularized low rank matrix approximation. *Journal of multivariate analysis* **99** 1015–1034.

[26] SOALE, A.-N. (2025). Detecting influential observations in single-index Fréchet regression. *Technometrics* **67** 311–322.

[27] SOALE, A.-N. and LUKMAN, A. (2025). Adaptive Influence Diagnostics in High-Dimensional Regression. *arXiv preprint arXiv:2510.15618*.

[28] SONG, Y., LI, Z. and FANG, M. (2022). Robust variable selection based on penalized composite quantile regression for high-dimensional single-index models. *Mathematics* **10** 2000.

[29] SZÉKELY, G. J. and RIZZO, M. L. (2014). Partial distance correlation with methods for dissimilarities. *The Annals of Statistics* **42** 2382 – 2412. https://doi.org/10.1214/14-AOS1255

[30] TIAN, Z. and QIU, P. (2024). Generalized single index modeling of longitudinal data with multiple binary responses. *Statistics in Medicine* **43** 3578–3594.

[31] TIBSHIRANI, R. (1996). Regression Shrinkage and Selection via the Lasso. *Journal of the Royal Statistical Society: Series B* **58** 267–288.

[32] WANG, Q. and YIN, X. (2008). A nonlinear multi-dimensional variable selection method for high dimensional data: Sparse MAVE. *Computational Statistics & Data Analysis* **52** 4512–4520.

[33] XIA, Y., TONG, H., LI, W. K. and ZHU, L.-X. (2002). An adaptive estimation of dimension reduction space. *Journal of the Royal Statistical Society Series B: Statistical Methodology* **64** 363–410.

[34] YIN, X., LI, B. and COOK, R. D. (2008). Successive direction extraction for estimating the central subspace in a multiple-index regression. *Journal of Multivariate Analysis* **99** 1733–1757.

[35] YÜZBAŞI, B., ARASHI, M. and EJAZ AHMED, S. (2020). Shrinkage estimation strategies in generalised ridge regression models: low/high-dimension regime. *International Statistical Review* **88** 229–251.

[36] ZHANG, C.-H. (2010). Nearly Unbiased Variable Selection Under Minimax Concave Penalty. *The Annals of Statistics* **38** 894–942.